

SUPPLEMENTAL MATERIAL “Deficiency in Vitamin B12 Biosynthesis and Propionate Catabolism Impairs Anaerobic Growth of Salmonella Dublin on Ethanolamine and 1,2-Propanediol”, by Martínez-Sanguin   et al, IAI 2026

Table S1. *S. Dublin* isolates used in this study for genome sequence analysis

Strain	Sequence Read Archive	BioSample	Source
SDu1	ERS022673	SAMEA811684	human blood isolate
SDu2	ERS022674	SAMEA811680	human blood isolate
SDu3	ERS022675	SAMEA811682	human blood isolate
SDu4	ERS022676	SAMEA811676	human blood isolate
SDu5	ERS022677	SAMEA811678	human feces isolate
SDu6	ERS022678	SAMEA811686	human feces isolate
SDu7	ERS022679	SAMEA811689	human feces isolate
SDu8	ERS354221	SAMEA2201818	human blood isolate
SDu9	ERS354222	SAMEA2201819	human blood isolate
SDu10	ERS354223	SAMEA2201820	human urine isolate
SDu73/96	ERS354224	SAMEA2201821	cattle
SDu76/96	ERS354225	SAMEA2201822	cattle
SDu73/04	ERS354226	SAMEA2201823	cattle
SDu75/04	ERS354227	SAMEA2201824	cattle

Table S2. Primers used in this study

Primer	Sequence (5'→3')	Application	Reference
cbiK_cl_F cbiK_cl_R	GGAGAATTCATGAAAAAGCGCTTCTGGTGGT ATTGGTACCTCATGCCGCCTCCTCTACCG	Amplification of <i>cbiK</i> for cloning	This work
prpE_cl_F prpE_cl_R	AAGGGTACCAGGAGAGCGTGATGTCTTTTAG GGCTCTAGACTATTCTCGATCGCCTGGC	Amplification of <i>prpE</i> for cloning	This work
prpR_cl_F prpR_cl_R	TCGTCTAGAAAAAGATGAGCCTACTGCGATGAC TCACTGCAGTTAATTATCCGACTGGTCTTTGGC	Amplification of <i>prpR</i> for cloning	This work
btuB_F btuB_R	GAATTAACCGCTGGATGGGG TTTCGCTACCACGTCAAACC	RTqPCR (btuB)	This work
btuC_F btuC_R	CCCTGCTATTGGCTGATGTG CCAGATAAACTGGTGCCC	RTqPCR (btuC)	This work
btuR_F btuR_R	GCTGCTTGATATGGTGGTCC CCTCTTCCAGCGGCAAAATAG	RTqPCR (btuR)	This work
cbiA_F cbiA_R	TGTATCTGGGGAGCACTCTG TACCCATTTTGTGTGACCG	RTqPCR (cbiA)	This work
cbiK_F cbiK_R	TTGCGTCCGTTATTCACCTCG TTTCTCCGTTGTCTGAGCG	RTqPCR (cbiK)	This work
cobC_F cobC_R	AAGATGCCGAAAGCTATGCC GTCGAGTCGCGAGATAAAGC	RTqPCR (cobC)	This work
cobD_F cobD_R	AATCTGGACTGCATTGAGCG CATTTCGCCAGTATCCAC	RTqPCR (cobD)	This work
cobU_F cobU_R	AAAACGATCCCGAACAGTGG CCCCATTCCCACCTCATTG	RTqPCR (cobU)	This work
cysG_F cysG_R	AGAAACTGGAATCGCTGCTG GCAAACCTGCTTTTACCCG	RTqPCR (cysG)	This work
eutB_F eutB_R	GCTGGAAACGATGGGCATTA GAAGAACAGTGACGGATCGC	RTqPCR (eutB)	[1]
eutM_F eutM_R	CATTGGCGAGCTGGTCTC AGCTGATCGGGAACACTTCT	RTqPCR (eutM)	[1]
eutR_F eutR_R	CGCTGGAGGTAAAAGGCAG ATGCAGGGTTTCAGGACTCT	RTqPCR (eutR)	This work
pduB_F pduB_R	GTTCATCGGTATTCTCGGC AATAATCAGCGAACCGTGCC	RTqPCR (pduB)	[1]
pduD_F pduD_R	CAGATTGGTAAAAACGCCGC GACTTTGCCTGGTACTTCGG	RTqPCR (pduD)	[1]
pocR_F pocR_R	GCTGGAAACGATGGGCATTA GAAGAACAGTGACGGATCGC	RTqPCR (pocR)	This work

In the cloning primers, restriction sites are shown in *italics*, putative Shine–Dalgarno sequences are underlined, and start codons are indicated in **bold**.

Anaerobic de novo pathway

Uroporphyrinogen III

↓ CysG

Precorrin-2

↓ CysG

syrohydrochlorin

↓ CbiK

Co-syrohydrochlorin

↓ CbiL

Co-Factor III

↓ CbiH

Co-precorrin-4

↓ CbiF

Co-precorrin-5A

↓ CbiG

Co-precorrin-5B

↓ CbiD

Co-precorrin-6A

↓ CbiJ

Co-precorrin-6B

↓ CbiET

Co-precorrin-8

→ CbiC → Cobyric acid → CbiA → Cob(II)yrinic acid a,c - diamide → CobR → Cob(II)yrinic acid a,c - diamide → CobA → Adenosylcobyrinic acid a,c - diamide → CbiP → Adenosylcobyrinic acid → CbiB → Adenosylcobinamide phosphate

Salvage pathway

Cobinamide

Outer membrane

Inner membrane

BtuB

BtuF

BtuC, BtuD

CobA

Adenosylcobinamide

Adenosylcobalamin

Adenosylcobalamin 5' - phosphate

CobC

CobS

CobV

Adenosylcobinamide - 5' - phosphate

GDP

Alfa-ribazole 5' - phosphate

5,6- dimethylbenzimidazole

CobU

Adenosylcobinamide phosphate

L- Threonine

Figure S2. CbiK protein alignment between *S. Dublin* and *S. Enteritidis*

SEn_PT4_CbiK	MKKALLVVSFGTSGYHDTCEKNIVACERDLAASCPDRDLRAFTSGMIIRKLQRDGDID	60
Sdu_CbiK	MKKRFWWSVLAPA-----IT	15
	*** : .: :	*
SEn_PT4_CbiK	TPLQALQLKAAQQYQDVAIQSLHIINGDEYEKIVREVQLLRPLFTRLTLGVPLLSSHNDY	120
Sdu_CbiK	TPVKK-----TLWRANVI-----W	29
	**:: .: * :	:
SEn_PT4_CbiK	VQLMQALRQQMPSLRQTEKVVFIMGHGASHHAFAAYACLDMMTAQRFPARVGAVESYPEV	180
Sdu_CbiK	RRLV-----PIATCFA-----PSLPE-	45
	:*: :*: * **	
SEn_PT4_CbiK	DILIDSLRDEGVTGVHLMPMLVLVAGDHAINMASDDGDSWKMRFNAAAGIPATPWLSGLGE	240
Sdu_CbiK	-----	45
SEn_PT4_CbiK	NPAIRAMFVAHLRQALNMTVEEAA	264
Sdu_CbiK	-----	45

Figure S2. Amino acid sequences alignment of CbiK from SEN PT4 and SDu strains. The alignment was performed using the ClustalO tool (<https://www.ebi.ac.uk/jdispatcher/msa/clustalo>).

Figure S3. 1,2-propanediol and propionate catabolic pathways in *S. Typhimurium*

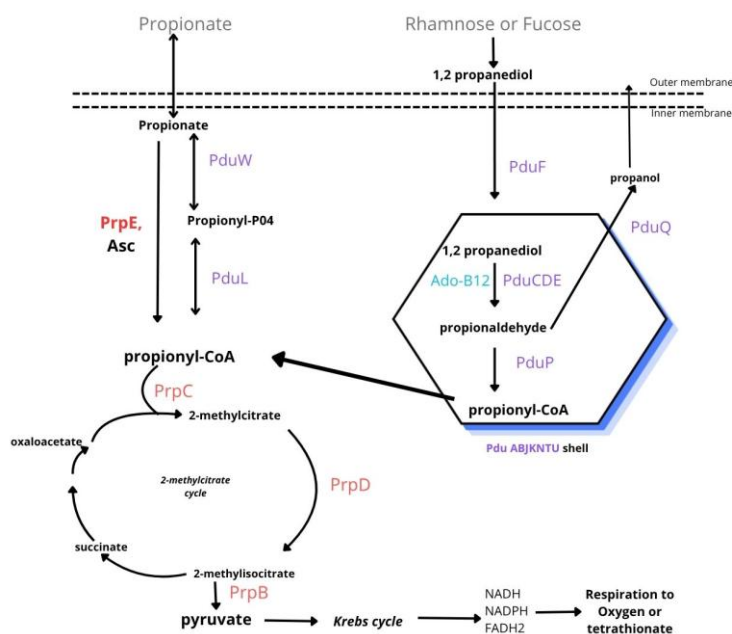


Figure S3. 1,2-propanediol and propionate catabolic pathways in *S. Typhimurium*. 1,2- propanediol catabolism is accomplished by enzymes encoded in the *pdu* operon (in purple). The polyhedral microcompartment is represented as an hexamer (the structural proteins component of the shell are shown below). Propionate catabolism is achieved by the *prp* operon (in pink). The common intermediate propionyl-CoA is converted into pyruvate through the action of PrpCDB enzymes, which can then be used as energy source through the TCA cycle, for final respiration using oxygen or as electron acceptor under aerobic or anaerobic conditions, respectively. The cofactor vitamin B12 is indicated in cyan (Based on Yoo et al.,2017, [3]).

Figure S4. *prpE* and PrpE alignment between *S. Dublin* and *S. Enteritidis*

A)

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PT4_prpE      ATGTCCTTTTAGCGAATTTTATCAGCGTTCATTAAACGAACCGGAGGCGTTCTGGGCCGAG 60
SDu_prpE      ATGTCCTTTTAGCGAATTTTATCAGCGTTCATTAAACGAACCGGAGGCGTTCTGGGCCGAG 60
*****

PT4_prpE      CAGGCCCCGACTATCGACTGGCGACAGCCGTTTACGCAGACGCTGGATCATAGCCGTCCA 120
SDu_prpE      CAGGCCCCGCGTATCGACTGGCGACAGCCGTTTACGCAGACGCTGGATCATAGCCGCCCA 120
*****

PT4_prpE      CCGTTTGGCCGCTGGTTTGTGCGCGGCACCACTAACTTATGTCATAACGCCGTGACCGC 180
SDu_prpE      CCGTTTGGCCGCTGGTTTGTGCGCGGCACCACTAACTTATGTCATAACGCCGTGACCGC 180
*****

PT4_prpE      TGGCTGGATAAACAGCCGGAGGCGCTGGCGCTGATTGCCGTCTCATCAGAGACCGATGAA 240
SDu_prpE      TGGCGGGATAAACAGCCGGAGGCGCTGGCGCTGATTGCCGTCTCATCAGAGACCGATGAA 240
****

PT4_prpE      GAGCGCACATTTACCTTCAGCCAGTTGCATGATGAAGTCAACGCTGTGGCCGCTATGCTG 300
SDu_prpE      GAGCGCACATTTACCTTCAGCCAGTTGCATGATGAAGTCAACATGTGGCCGCCATGTTG 300
*****

PT4_prpE      CTGTCGCTGGGCGTGCAGCGTGGCGATCGCGTATTGGTCTATATGCCGATGATTGCCGAA 360
SDu_prpE      CTGTCGCTGGGCGTGCAGCGTGGCGATCGCGTATTGGTCTATATGCCGATGATTGCCGAA 360
*****

PT4_prpE      GCGCAGATAACCCTGCTGGCCTGTGCGCGCATTTGGCGCGATCCATTGCGTGGTCTTTGGC 420
SDu_prpE      GCGCAGATAACCCTGCTGGCCTGTGCGCGCATTTGGCGCGATCCATTGCGTGGTCTTTGGC 420
*****

PT4_prpE      GGTTTTGCCTCGCACAGCGTGGCGGCGCGATTGACGATGCCAGACCGCGCTGATTGTG 480
SDu_prpE      GGTTTTGCCTCGCACAGCGTGGCGGCGCGATTGACGATGCCAGACCGCGCTGATTGTG 480
*****

PT4_prpE      TCGGCGGATGCCGGAGCGCGGGGCGGTAAATCCTGCCGTATAAAAAGCTGCTTGATGAC 540
SDu_prpE      TCGGCGGATGCCGGAGCGCGGGGCGGTAAATCCTGCCGTATAAAAAGCTGCTTGATGAC 540
*****

PT4_prpE      GCTATTGCGCAGGCGCAGCATCAGCCGAAACAGTTCTGCTGGTGGACAGAGGGCTGGCG 600
SDu_prpE      GCTATTGCGCAGGCGCAGCATCAGCCGAAACAGTTCTGCTGGTGGACAGAGGGCTGGCG 600
*****

PT4_prpE      AAAATGGCGTGGGTGGATGGGCGCGATCTGGATTTTGCCACGTTGCGCCAGCAGCATCTC 660
SDu_prpE      AAAATGGCGTGGGTGGATGGGCGCGATCTGGATTTTGCCACGTTGCGCCAGCAGCATCTC 660
*****

PT4_prpE      GGCGCGAGCGTGCCGGTGGCGTGGCTGGAATCCAATGAAACCTCGTGCAATCTTTACACC 720
SDu_prpE      GGCGCGAGCGTGCCGGTGGCGTGGCTGGAATCCAATGAAACCTCGTGCAATCTTTACACC 720
*****

PT4_prpE      TCCGGCACTACCGGCAAACCGAAAGGCGTCCAGCGCGACGTCGCGGTTATGCGGTGGCG 780
SDu_prpE      TCCGGCACTACCGGCAAACCGAAAGGCGTCCAGCGCGACGTCGCGGTTATGCGGTGGCG 780
*****

PT4_prpE      CTGGCAACCTCGATGGACACCATTTTTGGCGGCAAGGCGGGCGGCGTATTCTTTTGGCGCA 840
SDu_prpE      CTGGCAACCTCGATGGACACCATTTTTGGCGGCAAGGCGGGCGGCGTATTCTTTTGGCGCA 840
*****

PT4_prpE      TCGGATATCGGCTGGGTGCTCGGCCACTCCTATATCGTTTACGCGCCGCTGCTGGCAGGC 900
SDu_prpE      TCGGATATCGGCTGGGTGCTCGGCCACTCCTATATCGTTTACGCGCCGCTGCTGGCAGGC 900
*****

PT4_prpE      ATGGCGACTATTGTTTACGAAGGACTGCCGACGTACCCGGACTGCGGGGTCTGGTGAAA 960
SDu_prpE      ATGGCGACTATTGTTTACGAAGGACTGCCGACGTACCCGGACTGCGGGGTCTGGTGAAA 960
*****

PT4_prpE      ATTGTCGAGAAATACCAGGTTAACCGGATGTTTTCCGCCCCGACCGCGATTGCGGTGCTG 1020
SDu_prpE      ATTGTCGAGAAATACCAGGTTAACCGGATGTTTTCCGCCCCGACCGCGATTGCGGTGCTG 1020
*****

PT4_prpE      AAAAAATTCCCGACGGCGCAAATCCGCAATCACGATCTCTCCTCGCTGGAGGCGCTTTAT 1080
SDu_prpE      AAAAAATTCCCGACGGCGCAAATCCGCAATCACGATCTCTCCTCGCTGGAGGCGCTTTAT 1080
*****

PT4_prpE      CTGGCCGGTGAGCCGCTGGACGAGCCGACGGCCAGTTGGGTGACGGAGACGCTGGGCGTA 1140

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SDu_prpE	CTGGCCGGTGAGCCGCTGGACGAGCCGACGGCCAGTTGGGTGACGGAGACGCTGGGTGTG	1140

PT4_prpE	CCGGTCATCGACAATTATTGGCAGACGGAGTCCGGCTGGCCGATCATGGCGCTGGCCCGC	1200
SDu_prpE	CCGGTTATCGACAAATTATTGGCAGACGGAGTCCGGCTGGCCGATCATGGCGCTGGCCCGC	1200

PT4_prpE	GCGCTGGACGACAGGCCGTCGCGTCTGGGAAGTCCCGGCGTGCCGATGTACGGTTATAAC	1260
SDu_prpE	GCGCTGGACGACAGGCCGTCGCGTCTGGGAAGTCCCGGCGTGCCGATGTACGGTTATAAC	1260

PT4_prpE	GTCCAGCTACTCAATGAAGTCACCGGCGAGCCCTGCGGCATAAATGAAAAAGGCATGCTG	1320
SDu_prpE	GTCCAGCTACTCAATGAAGTCACCGGCGAGCCCTGCGGCATAAATGAAAAAGGGATGCTG	1320

PT4_prpE	GTGATCGAAGGGCCGCTGCCGCCGGGCTGTATTTCAGACTATTTGGGGCGACGATGCGCGT	1380
SDu_prpE	GTGATCGAAGGGCCGCTGCCGCCGGGCTGTATTTCAGACTATCTGGGGCGACGATGCGCGT	1380

PT4_prpE	TTTGTGAAGACTTACTGGTCGCTGTTTACCCGTCAGGTTTATGCCACCTTCGACTGGGGA	1440
SDu_prpE	TTTGTGAAGACTTACTGGTCGCTGTTTACCCGTCAGGTTTATGCCACCTTCGACTGGGGA	1440

PT4_prpE	ATCCGCGACGCCGAGGGGTATTACTTTTATCTGGGCCGTACCGATGATGTGATTAATATT	1500
SDu_prpE	ATCCGCGACGCCGAGGGGTATTACTTTTATCTGGGCCGTACCGATGATGTGATTAATATT	1500

PT4_prpE	GCGGGTCATCGGCTGGGGACGCGAGAAATAGAAGAAAGTATCTCCAGCTACCCGAACGTA	1560
SDu_prpE	GCGGGTCATCGGCTGGGGACGCGAGAAATAGAAGAAAGTATCTCCAGCTACCCGAACGTA	1560

PT4_prpE	GCGGAAGTGCGGTAGTGGGGATAAAAGACGCTCTGAAAGGGCAGGTGGCGGTGGCGTTT	1620
SDu_prpE	GCGGAAGTGCGGTAGTGGGGATAAAAGACGCTCTGAAAGGGCAGGTAGCGGTGGCGTTT	1620

PT4_prpE	GTCATTCCGAAGCAGAGCGATACGCTGGCGGATCGCGAGGCGGCGCGACGAGGAAAAC	1680
SDu_prpE	GTCATTCCGAAGCAGAGCGATACGCTGGCGGATCGCGAGGCGGCGCGACGAGGAAAAC	1680

PT4_prpE	GCGATTATGGCGCTGGTAGACAATCAGATTGGTCACTTTGGTCGTCCGGCGCATGTCTGG	1740
SDu_prpE	GCGATTATGGCGCTGGTAGACAATCAGATTGGTCACTTTGGTCGTCCGGCGCATGTCTGG	1740

PT4_prpE	TTTGTCCTCGCAGCTCCCCAAAACGCGTTCCGGAAAGATGCTTCGCCGCACGATCCAGGCG	1800
SDu_prpE	TTTGTTTCGCAGCTTCCCAAACGCGTTCCGGAAAGATGCTTCGCCGCACGATCTAG---	1797

PT4_prpE	ATCTGCGAAGGCCGCGATCCGGGCGATCTGACAACCATGTGACGATCCCGCGTCGTTGCAG	1860
SDu_prpE	-----	1797
PT4_prpE	CAAAATTCGCCAGGCGATCGAGGAATAG	1887
SDu_prpE	-----	1797

B)

SDu_PrpE	MSFSEFYQRSINEPEAFWAEQARRIDWRQPFTQTLDHSPFPFARWFCGGTTNLCHNAVDR	60
PT4_PrpE	MSFSEFYQRSINEPEAFWAEQARRIDWRQPFTQTLDHSPFPFARWFCGGTTNLCHNAVDR	60

SDu_PrpE	WRDKQPEALALIAVSSETDEERTFTFSQLHDEVNIVAAMLLSLGVQRGDRVLVYMPMIAE	120
PT4_PrpE	WLDKQPEALALIAVSSETDEERTFTFSQLHDEVNAAMLLSLGVQRGDRVLVYMPMIAE	120
	* *****	
SDu_PrpE	AQITLLACARIGAIHSVVFGGFASHSVAARIDDPALIVSADAGARGGKILPYKKLLDD	180
PT4_PrpE	AQITLLACARIGAIHSVVFGGFASHSVAARIDDPALIVSADAGARGGKILPYKKLLDD	180

SDu_PrpE	AIAQAQHQPkhvLLVDRGLAKMAWVDGRDLDFATLRQQHLGASVPVWLESNETSCILYT	240
PT4_PrpE	AIAQAQHQPkhvLLVDRGLAKMAWVDGRDLDFATLRQQHLGASVPVWLESNETSCILYT	240

SDu_PrpE	SGTTGKPKGVQRDVGGYAVALATSMDTIFGGKAGGVFFCASDIGWVVGHSYIVYAPLLAG	300
PT4_PrpE	SGTTGKPKGVQRDVGGYAVALATSMDTIFGGKAGGVFFCASDIGWVVGHSYIVYAPLLAG	300

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SDu_PrpE    MATIVYEGLPTYPCGVVWKIVEKYQVNRMFSAPTAIRVLKKFPTAQIRNHDLSSEALY    360
PT4_PrpE    MATIVYEGLPTYPCGVVWKIVEKYQVNRMFSAPTAIRVLKKFPTAQIRNHDLSSEALY    360
*****

SDu_PrpE    LAGEPLDEPTASWVTETLGVPVIDNYWQTESGWPIMALARALDDRPSRLGSPGVPMYGYN    420
PT4_PrpE    LAGEPLDEPTASWVTETLGVPVIDNYWQTESGWPIMALARALDDRPSRLGSPGVPMYGYN    420
*****

SDu_PrpE    VQLLNEVTGEPCGINEKGMLVIEGPLPPGCIQTIWGDDARFVKTYWSLFRQVYATFDWG    480
PT4_PrpE    VQLLNEVTGEPCGINEKGMLVIEGPLPPGCIQTIWGDDARFVKTYWSLFRQVYATFDWG    480
*****

SDu_PrpE    IRDAEGYYFILGRITDDVINIAGHRLGTREIEESSISSYPNVAEVAVVGIKDALKGQVAVAF    540
PT4_PrpE    IRDAEGYYFILGRITDDVINIAGHRLGTREIEESSISSYPNVAEVAVVGIKDALKGQVAVAF    540
*****

SDu_PrpE    VIPKQSDTLADREAARDEENAIMALVDNQIGHFGRPAHVWFVSQLPKTRSGKMLRRTI--    598
PT4_PrpE    VIPKQSDTLADREAARDEENAIMALVDNQIGHFGRPAHVWFVSQLPKTRSGKMLRRTIQA    600
*****

SDu_PrpE    ----- 598
PT4_PrpE    ICEGRDPGDLTTIDDPASLQQIRQAIEE 628

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Figure S4. A) Alignment of *prpE* gene sequences from SEn PT4 and *S. Dublin* (SDu) isolates used in this study. The start codons are highlighted in grey. The premature stop codon in serovar Dublin sequence is highlighted in green. *S. Enteritidis* stop codon is highlighted in grey. B) Amino acid sequences alignment of PrpE from SEn PT4 and SDu isolates used in this study. The alignments were performed using the ClustalO tool (<https://www.ebi.ac.uk/jdispatcher/msa/clustalo>).

Figure S5. *prpR* and PrpR alignment between *S. Dublin* and *S. Enteritidis*

A)

prpR_SDu	ATGACGACTGCCACAGCGCTCCGCGCGATAATAGCGATAAACCGGTGATCTGGACGGTC	60
prpR_PT4	ATGACGACTGCTCACAGCGCTCCGCGCGATAATAGCGATAAACCGGTGATCTGGACGGTC	60

prpR_SDu	TCCGTAACGCGTCTGTTCGAACTGTTTCGGGATATCAGCCTGGAATTCGATCATCTGGCG	120
prpR_PT4	TCCGTAACGCGTCTGTTCGAACTGTTTCGGGATATCAGTCTGGAATTCGATCATCTGGCG	120

prpR_SDu	ACCATCACGCTTATCAACTCGGCTTTGAAAAGGCGGTGACCTACATTTCGCAAAAACTG	180
prpR_PT4	ACCATCACGCTTATCAACTCGGCTTTGAAAAGGCGGTGACCTACATTTCGCAAAAACTG	180

prpR_SDu	GCGACCGAGCGCTGCGACGCGATTATCGCGCGGGTTCGAATGGGGCCTATTTAAAAAGC	240
prpR_PT4	GCGACCGAGCGCTGCGACGCGATTATCGCGCGGGTTCGAATGGGGCCTATTTAAAAAGC	240

prpR_SDu	CGCCTGTCAATACCGGTGATCCTCATCAAAACCCAGCGGATTCGATGTATTACAGGCGCTG	300
prpR_PT4	CGCCTGTCAATACCGGTGATCCTCATCAAGCCCAGCGGATTCGATGTATTACAGGCGCTG	300

prpR_SDu	GCGAAAGCGGAAAGCTCACCTCGTCTATCGGTATCGTGACCTATCAGGAGACCATTCCG	360
prpR_PT4	GCGAAAGCGGAAAGCTCACCTCGTCTATCGGTATCGTGACCTATCAGGAGACCATTCCG	360

prpR_SDu	GCGTTACTTGCCTTTCAGAAAACGTTTCACCTCCGTCTTGAACAGCGAAGCTATGTCACC	420
prpR_PT4	GCTTTACTTGCCTTTCAGAAAACGTTTCACCTCCGTCTTGAACAGCGAAGCTATGTCACC	420
** *****		
prpR_SDu	GAAGAGGACGCGCGGGGCAAATTAACGAACTTAAGGCCAACGGTATTGAGGCCGTCGTC	480
prpR_PT4	GAAGAGGACGCGCGGGGCAAATTAACGAACTTAAGGCCAACGGTATTGAGGCCGTCGTC	480

prpR_SDu	GGCGCGG-----	487
prpR_PT4	GGCGCGGGATTAATTACCGATCTGGCGGAAGAAGCGGGAATGACCGCCATCTTTATTTAT	540

prpR_SDu	-----GATATGACCCGTCTGACACGG	508
prpR_PT4	TCCGCGGCGACCGTTCGTCAGGCTTTTCATGATGCGCTGGATATGACCCGTCTGACACGG	600

prpR_SDu	CGCCAGCGCGTGGATTACCCATCCGGCAAGGGATTACAAACCCGGTATGAACTGGGCGAT	568
prpR_PT4	CGCCAGCGCGTGGATTACTCATCCGGCAAGGGATTACAAACCCGGTATGAACTGGGCGAT	660

prpR_SDu	ATACGCGGCCAGTCGCCGCAAATGAGCAGCTCCGCCAGACGATTACGCTCTATGCCCGC	628
prpR_PT4	ATACGCGGCCAGTCGCCGCAAATGGAGCAGCTCCGCCAGACGATTACGCTCTATGCCCGC	720

prpR_SDu	TCCCGTGCGGCAGTGCTGATTACAGGGGAAACAGGGACCGGAAAAGAGCTGGCGGCGCAG	688
prpR_PT4	TCCCGTGCGGCAGTGCTGATTACAGGGGAAACAGGGACCGGAAAAGAGCTGGCGGCGCAG	780

prpR_SDu	GCGATTACACGACGTTTTTTCACCGCCAGCCTCACCGTCAGAATAAGCCATCCCCTCCC	748
prpR_PT4	GCGATTACACGACGTTTTTTCACCGCCAGCCTCACCGTCAGAATAAGCCATCCCCTCCC	840

prpR_SDu	TTTGTCGCCGTCAATTGCGGCGCGATTACCGAGTCGTTGCTGGAAGCGGAACTGTTTGGT	808
prpR_PT4	TTTGTCGCCGTCAATTGCGGCGCGATTACCGAGTCGTTGCTGGAAGCGGAACTGTTTGGT	900

prpR_SDu	TATGAAGAGGGCGCGTTTACCGGTTACACGACGAGGAGGCCGGGCGGGCTGTTTGAAATC	868
prpR_PT4	TATGAAGAGGGTGCCTTTACCGGTTACACGACGAGGAGGCCGGGCGGGCTGTTTGAAATC	960

prpR_SDu	GCACACGGCGGCACGCTGTTTCTGGATGAAATCGGCGAAATGCCCTTGCCGTTACAAACC	928
prpR_PT4	GCACACGGCGGCACGCTGCTTCTGGATGAAATCGGCGAAATGCCCTTGCCGTTACAAACC	1020

prpR_SDu	CGACTTTTGCGCGTACTGGAGGAAAAAGCCGTCCTCGCGTCGGCGGACACCAGCCGATC	988
prpR_PT4	CGACTTTTGCGCGTACTGGAGGAAAAAGCCGTCCTCGCGTCGGCGGACACCAGCCGATC	1080

prpR_SDu	CCGGTGGATGTCCGGGTGATCAGCGCTACGCATTGCGATCTGGATCGGGAGATAATGCAA	1048
prpR_PT4	CCGGTGGATGTCCGGGTGATCAGCGCTACGCATTGCGATCTGGATCGGGAGATAATGCAA	1140

prpR_SDu	GGACGTTTTCGCCCCGATCTCTTTTATCGCCTGAGTATTCTGCGTCTGACGCTTCCTCCT	1108
prpR_PT4	GGACGTTTTCGCCCCGATCTCTTTTATCGCCTGAGTATTCTGCGTCTGACGCTTCCTCCT	1200

prpR_SDu	TTGCGCGAGCGGCAGGCTGATATTTTACCCTGCGGAAAGCTTTTAAAACAGTCGCTG	1168
prpR_PT4	TTGCGCGAGCGGCAGGCTGATATTTTACCCTGCGGAAAGCTTTTAAAACAGTCGCTG	1260

prpR_SDu	GCGGCGATGGAAATCCGTTTACCGAATCGATACGTCATGGATTGACACAGTGTAGCCG	1228
prpR_PT4	GCGGCAATGGAAATCCGTTTACCGAATCGATACGTCATGGATTGACACAGTGTAGCCG	1320

prpR_SDu	CTTTTGCTGGCCTGGCGCTGGCCCGGTAATATTCGCGAACTGCGTAATATGATGGAGCGC	1288
prpR_PT4	CTTTTGCTGGCCTGGCGCTGGCCCGGTAATATTCGCGAACTGCGTAATATGATGGAGCGC	1380

prpR_SDu	CTGGCGCTTTTTTTAAGCGTCGATCCCGCGCCAACGCTGGACAGGCAATTTATGCGGCAG	1348
prpR_PT4	CTGGCGCTTTTTTTAAGCGTCGATCCCGCGCCAACGCTGGACAGGCAATTTATGCGGCAG	1440

prpR_SDu	TTATTACCTGAGCTTATGGTGAACACAGCAGAAGTACGCTTCAACCGTGGATGCGCAC	1408
prpR_PT4	TTATTACCTGAGCTTATGGTGAACACAGCAGAAGTACGCTTCAACCATGGATGCGCAC	1500

prpR_SDu	ACGTTACAGGATGTACTGGCGCGCTTTAAGGGCGATAAGAGCGCCACGGCGCGTTATCTG	1468
prpR_PT4	ACGTTACAGGATGTACTGGCGCGCTTTAAGGGCGATAAGAGCGCGCGCGCGTTATCTG	1560

prpR_SDu	GGGATTAGCCGCACCACTCTGTGGCGTCGTTTAAAGCTGGTGCCAAAGACCAGTCGGAT	1528
prpR_PT4	GGGATTAGCCGCACCACTCTGTGGCGTCGTTTAAAGCAGGTGCCAAAGACCAGTCGGAT	1620

prpR_SDu	AATTAA 1534	
prpR_PT4	AATTAA 1626	

B)

PrpR_1_SDu	MTTAHSAPRDNNDKPVIVTVSVTRLFELFRDISLEFDHLATITPIQLGFEKAVTYIRKKL	60
PrpR_2_SDu	-----	0
PrpR_PT4	MTTAHSAPRDNNDKPVIVTVSVTRLFELFRDISLEFDHLATITPIQLGFEKAVTYIRKKL	60
PrpR_1_SDu	ATERCDAIIAAGSNGAYLKSRLSIPVILIKPSGFDVLQALAKAGKLTSSIGIVTYQETIP	120
PrpR_2_SDu	-----	0
PrpR_PT4	ATERCDAIIAAGSNGAYLKSRLSIPVILIKPSGFDVLQALAKAGKLTSSIGIVTYQETIP	120
PrpR_1_SDu	ALLAFQKTFHLRLEQRSYVTEEDARGQINELKANGIEAVVGAGYDPSDTAPARGL-----	175
PrpR_2_SDu	-----	0
PrpR_PT4	ALLAFQKTFHLRLEQRSYVTEEDARGQINELKANGIEAVVGAGLITDLAEEAGMTAIFY	180

PrpR_1_SDu	---PIRQGITNPV-----	185
PrpR_2_SDu	-----MEQLRQTITLYAR	13
PrpR_PT4	SAATVRQAFHDALDMTRLTRRQVRVDYSSGKGLQTRYELGDIRGQSPQMEQLRQTITLYAR	240
PrpR_1_SDu	-----	185
PrpR_2_SDu	SRAAVLIQGETGTGKELAAQAIHQTFHHRQPHRQNKPSPPFVAVNCGAITESLLEAELFG	73
PrpR_PT4	SRAAVLIQGETGTGKELAAQAIHQTFHHRQPHRQNKPSPPFVAVNCGAITESLLEAELFG	300
PrpR_1_SDu	-----	185
PrpR_2_SDu	YEEGAFTGSRRGGRAGLFEIAHGGTFLFLDEIGEMPLPLQTRLLRVLEEKAVTRVGGHQPI	133
PrpR_PT4	YEEGAFTGSRRGGRAGLFEIAHGGTLLFLDEIGEMPLPLQTRLLRVLEEKAVTRVGGHQPI	360
PrpR_1_SDu	-----	185
PrpR_2_SDu	PVDVRVISATHCDLDREIMQGRFRPDLFYRLSILRLTLPPLRERQADILPLAESFLKQSL	193
PrpR_PT4	PVDVRVISATHCDLDREIMQGRFRPDLFYRLSILRLTLPPLRERQADILPLAESFLKQSL	420
PrpR_1_SDu	-----	185
PrpR_2_SDu	AAMEIPFTESIRHGLTQCQPLLAWRWPGNIRELRNMMERLALFLSVDPAPTLDQRQFMRQ	253
PrpR_PT4	AAMEIPFTESIRHGLTQCQPLLAWRWPGNIRELRNMMERLALFLSVDPAPTLDQRQFMRQ	480
PrpR_1_SDu	-----	185
PrpR_2_SDu	LLPELMVNTAELTPSTVDAHTLQDVLARFKGDKSATARYLGISRTTLWRRLKAGAKDQSD	313
PrpR_PT4	LLPELMVNTAELTPSTMDAHTLQDVLARFKGDKTAAARYLGISRTTLWRRLKAGAKDQSD	540
PrpR_1_SDu	-	185
PrpR_2_SDu	N	314
PrpR_PT4	N	541

Figure S5. A) Alignment of *prpR* gene sequences from SEn PT4 and SDu isolates used in this study. The premature stop codon in the serovar Dublin sequence is highlighted in green. The start and end codons in SEn are highlighted in grey. The potential start codon after the deletion and frameshift in the serovar Dublin is highlighted in yellow. B) Alignment of PrpR amino acid sequences from strains SEn PT4 and SDu isolates used in this study. The notation prpR_1 denotes reading frame one and prpR_2 denotes reading frame two, after the deletion. The start methionines are highlighted in grey. The alignments were performed using the ClustalO tool (<https://www.ebi.ac.uk/jdispatcher/msa/clustalo>).